CCTV EXAM ANALYTICS · CLIENT ONBOARDING

Exam Onboarding Guide

The information, data files and file formats required to set up and conduct an examination under CCTV analytics monitoring, for both RTSP and HLS camera feeds.

PREPARED BY	APPLIES TO	FORMATS VALID AS OF	FILES REQUIRED
CDIS	HLS examinations	October 2026	7 data files, plus user list

1 Overview

The platform monitors examination centre cameras during each shift and detects events such as a mobile phone in a lab, a crowd at the entry gate or unauthorised presence in the server room. Each event is saved with an annotated snapshot. Every detection is reviewed by a CDIS analyst before it is released, so only verified alerts reach the client dashboard and the observers' mobile phones.

Processing stages

1. **Connectivity check**
   Each camera is opened over its RTSP or HLS link. Cameras that cannot be reached are marked Not Accessible and excluded from analysis.
2. **View classification**
   An AI model assigns each accessible camera a view type (Computer Lab, Server Room, Entry, Outside and so on), so that each analytic runs only on the camera types it applies to.
3. **Detection during the shift**
   Only the analytics scheduled for the shift run, on the camera types and within the time windows defined in the analytics plan.
4. **Analyst review**
   Detections are verified by CDIS analysts. Verified detections are released to the client Detections page.
5. **Client action**
   The client accepts a valid alert, which triggers an SMS to the designated observers, or removes a false alert. No detection is deleted, and removals remain available for review.
6. **Reporting**
   A per-shift Excel report lists camera status and every detection with a link to its image.

RTSP and HLS examinations compared

	RTSP EXAMINATION	HLS EXAMINATION
Video source	A DVR server. CDIS signs in with an account provided by the client and requests each camera by Device ID and Camera No.	A web link per camera that returns an `.m3u8` playlist, normally published by a third-party CCTV portal.
Camera list	One file for the whole examination.	One sheet per shift, for each exam date. A camera is listed in every shift in which it is used.
Camera identifier	Device ID + Camera No + TC Code	TC Code + `ch_no`, within a shift and date
Stream availability	The DVR session must open successfully during the shift window.	The link must remain valid throughout the shift. A link that works only inside a limited time window must be live during CDIS checks and the shift.
Examination type	An examination is created as either RTSP or HLS, and this cannot be changed afterwards. An examination that uses both is set up as two examinations.	

2 Deliverables

The TC code links all files together. The schedules, camera list, server room limits and SMS recipients are matched by TC code alone, so each TC code must be written identically in every file. Blank templates for Files 2 to 6 and for the RTSP and HLS camera lists are supplied by CDIS with this guide.

REF.	ITEM	RTSP	HLS	FORMAT	PURPOSE
File 1	Exam details	REQUIRED	REQUIRED	Email or form	Exam name, body, dates, shifts and link type
File 2	Date and shift schedule	REQUIRED	REQUIRED	Excel	Dates and shifts on which the examination runs
File 3	Centre-wise schedule	REQUIRED	RECOMMENDED	Excel	Centres participating in each date and shift
File 4	Camera list	REQUIRED one file	REQUIRED one sheet per shift	Excel	Every camera to be monitored
File 5	Server room limits	IF APPLICABLE	IF APPLICABLE	Excel	Maximum permitted headcount per server room
File 6	SMS recipients and templates	IF SMS USED	IF SMS USED	Excel and template details	Observer numbers and DLT-approved message templates
File 7	Analytics plan (SOP)	REQUIRED	REQUIRED	SOP table (Excel)	Analytics, camera types and time windows
–	Portal users	REQUIRED	REQUIRED	List	Name, email, mobile and company of each user

3 Setup timeline

The sequence below reflects the recommended lead times. Exact dates are confirmed with your CDIS coordinator.

Before the first examination day

- Submit Files 1 to 7 and the user list. Every upload is validated and, if incorrect, rejected with row-level messages, so allow time for corrections.
- CDIS creates the examination and shifts, loads the schedules and cameras, and registers the SMS templates.
- CDIS runs a scene classification and a connectivity check, then shares the list of cameras that are Not Accessible or Uncertain so that they can be corrected before the examination.
- Users register and are verified (see Section 6).
- CDIS schedules the analytics from the submitted plan and confirms the timings with the client.

On each examination day

- Approximately two hours before a shift: scene classification and connectivity check of that shift's cameras.
- Before the shift starts: tamper, inversion and entry/exit checks begin.
- During the shift: in-room analytics (phone, proximity, invigilator, crowd, server room) run for the duration of the shift.
- The client accepts or removes alerts as they arrive and downloads the shift report afterwards.
- If a schedule changes during the examination, the corrected file replaces the previous one and takes effect from the next launch of that shift's job. For camera list changes (new links, moved or removed cameras), inform CDIS: new cameras are added on upload, and changes to existing cameras are applied by CDIS.

4 File specifications

Each file must be submitted in exactly the format shown below. These formats are identical to the blank templates supplied by CDIS with this guide, and header names must match them character for character. Files that deviate are rejected at upload or need manual re-entry by CDIS. Columns marked with a green underline are read by the system; the remaining columns are informational and may be left empty.

Column required or read by the system

General requirements for all Excel files

- Accepted file types: `.xlsx`, `.xls` and `.ods`. Maximum size: **10 MB**.
- Only the **first sheet** of a workbook is read. If a workbook contains several tabs (for example SOP, Exam Schedule, Server Room Threshold), the required tab must be placed first or saved as a separate file.
- Row 1 holds the headers, except in File 2, whose template has the title `Exam Schedule` in row 1 and the headers in row 2.
- TC codes are text, written identically in every file (for example `50106`, with no trailing `.0`).
- All times must be in 24-hour format (`13:30`, not `01:30`).
- Schedule, threshold and recipient uploads display a preview first. Errors prevent saving and are listed by row; no upload is partially saved.

LAYOUTS THAT CANNOT BE PROCESSED

The layouts below cannot be read automatically. CDIS can transcribe them manually, but this adds time and the risk of transcription error. Please submit the layouts specified in this guide.

- **Dates across, shifts below.** A schedule laid out horizontally (`Date | 17th Sept | | | 18th Sept` followed by `Shifts | Shift 1 | Shift 2 | Shift 3`). Use one row per date instead (File 2).
- **Two header rows.** Threshold or schedule matrices with the date on one row and `S1 S2 S3` on the next. Combine them into a single header such as `17-9-26 Shift 1` .
- **Contact lists without a TC code.** A list such as `Sr No | Contact Number` . See File 6 for the two accepted ways to supply numbers.
- **Times without AM/PM or with a dot separator.** Examples: `01:30:00` meaning 1:30 PM, or `4.30 AM` . Write `13:30` and `04:30` .

File 1 Exam details

RTSP HLS

EXAM NAME Sample Combined Graduate Level Exam 2026	**EXAM BODY** Name of the body conducting the examination
LINK TYPE RTSP or HLS (cannot be changed later)	**EXAM PERIOD** Start date and end date
NON-EXAM DAYS Weekdays to skip (for example Sunday) and specific dates	**SHIFTS** Shift name, start time and end time in 24-hour format, for example Shift 1 10:00–11:00 and Shift 2 13:30–14:30

NOTE

The exam period, skipped dates and shifts can be read directly from the date and shift schedule (File 2), so they need not be supplied twice.

File 2 Date and shift schedule

RTSP HLS REQUIRED

One row per examination date and one column per shift, with **Yes** or **No** in each cell. A slot marked No, or a date that is absent from the sheet, means the examination does not run and nothing is monitored. This schedule always takes precedence over the centre-wise schedule.

`Exam Schedule (row 1 is the title; headers are in row 2)`

Sr.No.	Date	Shift 1 (10:00:00 to 11:00:00)	Shift 2 (13:30:00 to 14:30:00)	Shift 3 (17:00:00 to 18:00:00)
1	30-Sep-26	No	Yes	Yes
2	1-Oct-26	Yes	Yes	Yes
3	5-Oct-26	Yes	Yes	No

- Row 1 contains the title `Exam Schedule` . Row 2 contains the headers: `Sr.No.` , `Date` , and one `Shift N (HH:MM:SS to HH:MM:SS)` column per shift.
- Date format: `30-Sep-26` (day, three-letter month, two-digit year). Each date appears once.

- Cell values: exactly `Yes` or `No`.
- "Shift N" corresponds to the Nth shift of the examination, so the shift order in File 1 must agree with this sheet.
- The sheet must be the first tab of the workbook, or a separate file.

> **VERIFY AFTERNOON TIMES**
> Source documents frequently show `Shift 2 (01:30:00 to 02:30:00)` and `Shift 3 (05:00:00 to 06:00:00)` where 13:30 and 17:00 are intended. The system raises a warning for any time before 06:00; if uncorrected, the night-time hours would be stored as the shift. Enter `13:30:00` and `17:00:00`.

File 3 Centre-wise schedule

RTSP · REQUIRED HLS · RECOMMENDED

One row per centre and one column per **date and shift**, with YES or NO in each cell. In each slot only the centres marked YES are classified and monitored. A centre with cameras that is missing from this sheet is not monitored on any date. For HLS examinations the camera sheets already list cameras per shift and date; this schedule additionally excludes centres from the slots in which they do not participate.

Centre schedule

TC CODE	30-9-26 Shift 1	30-9-26 Shift 2	1-10-26 Shift 1	1-10-26 Shift 2
10001	NO	YES	YES	YES
10002	NO	NO	YES	NO

- Row 1 contains the headers: `TC CODE`, followed by one column per date and shift. No other columns are required.
- Date and shift header format: `d-m-yy Shift n`, with single spaces, for example `30-9-26 Shift 1`. Include one column for every date and shift of the examination; each combination appears once.
- Cell values: exactly `YES` or `NO`. An empty cell is treated as NO with a warning.
- One row per centre; each TC code appears once, and every centre that has cameras must be listed.

File 4 Camera list

RTSP HLS REQUIRED

HLS One sheet per shift, for each exam date

HLS Cameras · one sheet per shift (Shift 1)

S.No.	TC Code	Centre Name	City	State	LIS IF/ IDZ	LAN and Lab Name	Camera Name	CCTV URL	ch_no	Camera Category
1	10001	Sample Public School	Lucknow	Uttar Pradesh	LIS IF	LAN1-LAB 2,LAB 1	10001/ L-01/C01	https:// portal.example.com/cctv/stream? centre=…&camera=…	55198	Inside Lab
2	10001	Sample Public School	Lucknow	Uttar Pradesh		OUTSIDE_ROOM_CCTV	10001/ ENTRY EXIT/S23	https:// portal.example.com/cctv/stream? centre=…&camera=…	55226	Entry/ Exit
3	10001	Sample Public School	Lucknow	Uttar Pradesh		OUTSIDE_ROOM_CCTV	10001/ SVRM/S22	https:// portal.example.com/cctv/stream? centre=…&camera=…	55307	Server Room

- Row 1 contains the headers exactly as shown: `S.No.`, `TC Code`, `Centre Name`, `City`, `State`, `LIS IF/IDZ`, `LAN and Lab Name`, `Camera Name`, `CCTV URL`, `ch_no`, `Camera Category`.
- **CCTV URL** is the camera stream link from the client's CCTV portal, typically a portal page link (for example `https://portal.example.com/cctv/stream?centre=…&camera=…`) that resolves to an `.m3u8` playlist. **ch_no** is the unique channel key of the camera. A row missing either value is skipped.
- The **shift and date** are selected at the time of upload. Supply one sheet per shift (`Shift 1`, `Shift 2`, `Shift 3`) for each exam date. The centres and cameras may differ from shift to shift, and the date must be an examination date.
- **List each camera once per sheet.** `ch_no` must be unique within a sheet; for a repeated `ch_no` only the first row is kept and the remaining rows are discarded with a message.
- **Camera Category** must be exactly one of `Inside Lab`, `Outside Lab`, `Entry/Exit` or `Server Room`, and must be completed for every camera. Other spellings (for example `Outside`, `Outside Room` or `Entry/Exit Gate`) are accepted at upload but are not recognised, and the camera is then assigned by the AI label instead. Where an analytic is restricted to certain camera types, this column takes precedence over the AI label. In the analytics plan (File 7), the Camera Type `Outside` applies to cameras with the category `Outside Lab`; the other three names are the same in both files.
- **LAN and Lab Name** and **LIS IF/IDZ** are informational and may be blank for outside cameras (for example `OUTSIDE_ROOM_CCTV`). An optional extra `Classification` column overrides the AI label when present.
- Links must remain valid for the entire shift window. Short-lived or time-restricted links are the most common cause of cameras appearing as Not Accessible.
- Re-uploading a sheet for a shift and date that is already loaded adds new cameras but does **not replace the links** of existing ones. If links change, for example refreshed or time-limited links, inform CDIS so that the sheet for that shift and date can be replaced.

File 5 Server room limits

RTSP HLS IF APPLICABLE

The maximum number of people permitted in each centre's server room. An alert is raised when the number of people observed is **greater than** the limit.

`Server room threshold`

TC CODE	Maximum Server Room Threshold
10001	11
10002	8

- Row 1 contains the headers `TC CODE` and `Maximum Server Room Threshold`. No other columns are required. One row per centre.
- Fractional values are rounded to the nearest whole person (`7.75` becomes `8`), and a limit of 0 is skipped.
- Once this file is uploaded for the examination, a centre with no row receives **no server room alerts**, and limits from other examinations are not used. If the file is not supplied, limits previously held by CDIS for a TC code may apply, so submit this file whenever server room monitoring is required.

Limits that vary by date and shift

If the limit differs between sittings, replace `Maximum Server Room Threshold` with one column per date and shift, using the header format of File 3. Keep a single header row.

`Server room threshold per sitting (single header row)`

TC CODE	14-10-26 Shift 1	14-10-26 Shift 2	16-10-26 Shift 1
10001	4	4	NA
10002	5	5	0

- Cell values are whole numbers. `0`, `NA` or an empty cell means the centre is not scheduled in that slot.

> **THRESHOLD SEMANTICS**
> The alert is raised when the headcount is **greater than** the limit (a limit of 4 alerts at 5 people). If the SOP states "equal to or more than the limit", supply the highest permitted headcount as the limit, or inform CDIS so that both sides interpret the figure identically.

`File 6` SMS recipients and templates

RTSP HLS IF SMS USED

When the client accepts an alert on the dashboard, an SMS is sent. The recipients can be defined in either of two ways.

Option A · recipients per centre

Two columns, `TC Code` and `Mobile Number`. The SMS is sent to the numbers listed for the centre of the alert.

Option B · one list for all centres

A plain list of mobile numbers (for example `Sr No | Contact Number`) for a central team. CDIS enters it against the examination and every alert is sent to these numbers. **This list must not be uploaded as a per-centre file**: without a TC code column, the first column would be read as the TC code.

For each alert the recipients are resolved in this order: the centre's numbers (Option A), then the examination-wide list (Option B), then the default recipients registered for the client's company.

`Observer Numbers`

TC Code	Mobile Number
10001	9000000001
10001	9000000002
10002	9000000003

- Row 1 contains the headers exactly as shown: `TC Code` and `Mobile Number`.
- One number per row, repeating the TC code for each additional number. Numbers are 10-digit mobile numbers without country code, spaces or dashes.
- Each upload **replaces** the entire per-centre list, so every upload must contain the complete list.
- Alerts are delivered by SMS only; email lists are not used for alerts.
- SMS is available for: invigilator (including crowd below 5 with invigilator static), proximity, server room, intruder, entry crowd, phone, entry/exit and blind spot.
- An alert type is sent by SMS only after its DLT template has been registered for your organisation. Templates are held per client organisation, so all examinations of one client use the same templates.

SMS templates (DLT)

Indian telecom regulations require each message body to match a template approved on the DLT portal. For every alert to be sent by SMS, provide the following five items.

ITEM	EXAMPLE
Template name	Close Proximity
Sender ID	The approved 6-character header
DLT template ID	The ID issued on the DLT portal
Venue / entity name	As registered
Message	`Two or more people are in close proximity in ##camname## at ##timestamp## - SENDER`

- Variables that can be populated: `##camname##` (camera), `##timestamp##` (time) and, where relevant, `##pplcount##` (people counted) and `##ppl##` (limit). Use only these variables in the message body.
- Templates are required for: Intruder, Crowd at entry gate, Invigilator not moving, Close proximity, Server room threshold breach, Mobile phone, Entry/Exit movement and Blind spot. The wording must match the registered template exactly.

File 7 Analytics plan (SOP)

RTSP HLS REQUIRED

The client's SOP table serves as the analytics plan and may be submitted as it stands, subject to the points below. CDIS enters each row on the scheduling screen. The expected layout has one row per analytic and camera type, with a Start and End pair under each shift.

SOP · analytics plan (24-hour times)

Sr. No.	Analytics	Threshold for raising alert	Alert gap (seconds)	Camera Type	S1 Start	S1 End	S2 Start	S2 End	S3 Start	S3 End	SMS to observer
1	Intruder detection	Any person present in the defined window	120	Inside Lab	05:00	06:00	-	-	-	-	No
		Any person present in the defined window	120	Server Room	05:00	06:00	-	-	-	-	No
2	Crowd detection	10 or more people	120	Entry/ Exit	09:15	10:15	12:45	13:45	16:15	17:15	No
		10 or more people	120	Outside	09:16	10:16	12:46	13:46	16:16	17:16	Yes
3	Invigilator not moving	7 minutes standing still	420	Inside Lab	10:00	11:00	13:30	14:30	17:00	18:00	Yes
4	Close proximity	Two or more people too close	120	Inside Lab	10:00	11:00	13:30	14:30	17:00	18:00	Yes
5	Mobile phone detection	Any phone	120	Inside Lab	10:00	11:00	13:30	14:30	17:00	18:00	Yes
		Any phone	120	Server Room	10:01	10:59	13:31	14:29	17:01	17:59	Yes

SOP documents usually place "Shift 1 / Shift 2 / Shift 3" on a merged row above "Start time / End time". The grid above shows them combined in a single header for display.

- **Camera Type** must be one of `Inside Lab`, `Server Room`, `Entry/Exit` or `Outside`. For HLS examinations only these four may be used: each corresponds to a Camera Category in the camera sheet, with `Outside` covering cameras of category `Outside Lab`. For RTSP examinations, `Computer Lab`, `Inside Classroom`, `Strong Room` and `Biometric` may also be used. Use one row per analytic and camera type, as in the Crowd and Phone rows above, so that each row can carry its own time window and SMS setting.

- **Start and End per shift** are 24-hour times in the form `HH:MM`, with the End after the Start. A dash indicates that the analytic does not run in that shift. A window may begin before the examination starts (intruder checks commonly do) and may differ for each analytic.
- **Alert gap (seconds)** is the minimum interval between two alerts from the same camera for that analytic, entered as a whole number of seconds (for example `120` for 2 minutes). Leave it blank to use the platform default for the analytic.
- **SMS to observer** is `Yes` or `No` for each row. Yes is effective only for the analytics listed in File 6 and requires a template and recipients.
- Alert thresholds, such as "10 or more people", "7 minutes without movement" or "any phone", are applied by the platform and are recorded in the SOP for reference. The server room limit is the exception and is supplied in File 5.
- If the same analytic runs on different camera types within one shift, offset the start of the second row by one minute (see the Phone rows above).

USE 24-HOUR TIMES
12-hour times are the most frequent source of error in SOP documents. Shift 2 is often written as `01:30:00 to 02:30:00` and Shift 3 as `05:00:00 to 06:00:00` (meaning 13:30 and 17:00), or times appear as `4.30 AM` and `2.00 PM`. Convert every time to `HH:MM` in 24-hour form before submission.

5 Conducting the examination

Client dashboard

- **Dashboard**: examinations by status (Scheduled, Ongoing, Completed), cameras Accessible versus Not Accessible, detections Pending, Accepted and Removed, and the analytics running that day.
- **Cameras**: every camera with TC code, centre, category, classification and status, filterable by shift and exam date. "Open frame" displays the snapshot used for classification.
- **Detections**: live feed with the annotated snapshot, camera, centre, TC code, shift and time, filterable by date, type, centre or city.
- **Video Uploads**: images and videos organised by TC code.

Handling alerts

- **Accept** confirms that an alert is valid. The SMS to the designated observers is then sent automatically using the registered DLT template.
- **Remove** dismisses a false positive from the active feed. The detection remains on the Removed Detections page and can be restored.
- No detection is deleted. Accepted and removed decisions are reflected in the shift report.

Reports

- The per-shift Excel report contains cameras with channel ID and status, event counts, and a Detections List with Status (Accepted, Rejected or Not Verified) and a link to each image.
- Cameras of centres that do not participate in a slot are left out of that shift's camera counts, so they are not reported as offline.

Client responsibilities on the day

- Keep DVRs powered and stream links valid for the whole shift.
- Notify CDIS promptly of any camera relocation, centre cancellation or change in shift timing.
- Review Not Accessible cameras with the centre, since a camera that cannot be opened cannot be analysed.

6 Portal access

- Each user registers with name, email, mobile number and company, and verifies the account with a one-time password (OTP) sent by email or SMS.
- A CDIS administrator then verifies the account and assigns the client (viewer) role. Verification is performed manually, so users should register before the examination week.
- Client users can view only the examinations, cameras and detections of their own company. A user who works for several companies can hold several memberships under a single login.
- Please provide each user's full name, email address, mobile number and company, together with the examinations that user should be able to view.

7 Submission checklist

- [ ] Every TC code is written identically in all files, as text, with no trailing `.0`.
- [ ] Exam details are complete and the link type (RTSP or HLS) is correct. It cannot be changed later.
- [ ] The date and shift schedule has one row per exam date, is on the first tab (or in its own file) and uses 24-hour shift times.
- [ ] The centre-wise schedule contains YES or NO for every date and shift, and every centre with cameras is listed.
- [ ] HLS: each camera has a stream link, a unique `ch_no` (listed once) and a Camera Category, with one sheet per shift. Links remain valid for the whole shift.
- [ ] Every centre with a server room has a limit in the server room file.
- [ ] SMS numbers are 10-digit mobiles (with TC codes for per-centre recipients, or a single list for all), and each SMS template includes its sender ID and DLT template ID.
- [ ] The SOP lists, for each shift, the analytic, camera type, start and end times in 24-hour form, cool-down and SMS setting.
- [ ] Portal users are listed and have registered.

8 Upload error reference

MESSAGE	CAUSE AND RESOLUTION
Only Excel files (.xls, .xlsx) are allowed	Submit `.xlsx`, `.xls` or `.ods` files; CSV, PDF and images are not accepted.
The file is too large (max 10 MB) or File too large	Split the camera list into several files.
No "TC CODE" column found	The header row must contain `TC CODE` (or `TC Code`) within the first 12 rows.
No date + shift columns found	Headers must follow the pattern `30-9-26 Shift 1`.
"MAYBE" is not YES or NO	Schedule cells accept only YES or NO.
TC code … appears more than once	Schedule and limit files must have one row per centre.
The exam has no shift for: Shift 4	The sheet refers to a shift that the examination does not have. Add the shift to the examination first.
Dates outside the exam period (warning)	The sheet lists dates beyond the examination start or end date. Correct the dates or change the exam period.
Rows have no readable channel id	HLS: rename the channel column to `ch_no`.
Rows repeat a CH NO already listed in the same file	HLS: the same camera appears on several rows. Keep one row per `ch_no`.
No header with a "Date" column and "Shift n" columns found	The schedule is laid out horizontally, or it is not on the first tab. Use one row per date and place the tab first.
Neither a "Maximum Server Room Threshold" column nor date + shift columns were found	The date and the shift are on separate header rows. Combine them, for example `17-9-26 Shift 1`.
Some shift times are before 06:00 (warning)	A 12-hour time such as `01:30` was used where 13:30 is intended. Rewrite in 24-hour form.
No valid server address (not imported)	RTSP: the server column must contain a host such as `dvr01.example.com` or an rtsp/http URL.

Prepared by CDIS for client onboarding. Formats correspond to the uploads accepted by the platform as of October 2026. The layouts and common errors described are drawn from examination files received from clients; all sample centres, codes, numbers and addresses shown in this guide are fictitious.